

Number systems

COMPLETE LEARNING GUIDE

What You'll Learn

Number systems questions form the backbone of almost every placement aptitude test because the shortcuts they teach — divisibility rules, HCF/LCM, remainder cyclicity, unit-digit tricks — get reused inside percentage, time-speed-distance, and data interpretation problems later. Unlike topics that reward memorized formulas alone, number systems reward pattern recognition: recognizing that a huge power's unit digit repeats every 2-4 steps, or that a factorial's trailing zeros depend only on how many times 5 divides into it, turns an intractable-looking question into a 20-second calculation. This sub-module builds the core toolkit — classification and divisibility, HCF/LCM and their word-problem applications, remainder and cyclicity shortcuts, unit-digit and trailing-zero techniques, and base conversion — with every rule demonstrated on a real worked number so the technique, not just the definition, sticks. Because these problems appear in nearly every test as quick, calculator-free questions, speed and accuracy here directly raise your overall score.

1. Classification of Numbers and Divisibility Rules

Placement tests lean heavily on divisibility rules because they let you answer 'is this divisible by X' without doing the full division. The rules worth memorizing cold: divisible by 2 (last digit even), by 3 (digit sum divisible by 3), by 4 (last two digits divisible by 4), by 5 (ends in 0 or 5), by 8 (last three digits divisible by 8), by 9 (digit sum divisible by 9), and by 11 (alternating digit sum from the right divisible by 11, including 0).

- Check 396528 for divisibility by 4, 8, 9, and 11: last two digits 28 !' $28/4=7$, divisible by 4. Last three digits 528 !' $528/8=66$, divisible by 8. Digit sum $3+9+6+5+2+8=33$, NOT divisible by 9 (33 isn't a multiple of 9). Alternating sum from the right: $8-2+5-6+9-3=11$, divisible by 11 (11 itself counts).

2. HCF and LCM: Prime Factorization and Word Problems

HCF (highest common factor) is found by multiplying the lowest power of every common prime factor; LCM (lowest common multiple) is found by multiplying the highest power of every prime factor appearing in any of the numbers. The two are linked by the identity $HCF \times LCM = \text{product of the two numbers}$ — a fast way to check your work or find one value if the other three are known. LCM word problems (things repeating together, like bells or traffic lights) are extremely common.

- $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$. HCF = $2^2 \times 3 = 12$. LCM = $2^2 \times 3^2 \times 5 = 180$. Check: $12 \times 180 = 2160 = 36 \times 60$.

- Three bells ring every 12, 18, and 24 minutes. They rang together at 9:00 AM — when next? LCM(12,18,24): $12=2^2 \times 3$, $18=2 \times 3^2$, $24=2^3 \times 3$! LCM= $2^3 \times 3^2=72$ minutes. Next together ring: 10:12 AM.

3. Remainder Theorems and Cyclicity

When a power is too large to compute directly, find the cyclicity of the base's remainders modulo the divisor and reduce the exponent modulo the cycle length. This converts an unmanageable calculation like $2^p \pmod{7}$ into a small lookup.

- Find the remainder when 2^p is divided by 7. Powers of 2 mod 7 cycle: $2^1=2$, $2^2=4$, $2^3=1 \pmod{7}$ — cycle length 3. $100 \pmod{3} = 1$ (since 99 is divisible by 3). So $2^p \pmod{7} = 2^{1 \pmod{3}} = 2^1 \pmod{7} = 2$.

4. Unit Digit Shortcuts for Large Powers

Unit digits of powers cycle with a period of at most 4 (digits 4 and 9 cycle in 2, digits 2,3,7,8 cycle in 4, and 0,1,5,6 never change). Find the base's cycle, reduce the exponent modulo the cycle length (using the cycle length itself, not 0, when the remainder is 0), and read off the matching position.

- Find the unit digit of 7^{123} . Cycle of 7: $7^1=7$, $7^2=49(9)$, $7^3=343(3)$, $7^4=2401(1)$ — period 4. $123 \pmod{4} = 3$ (120 is divisible by 4). Third position in the cycle (7,9,3,1) is 3. Unit digit = 3.

5. Factorials and Trailing Zeros

Trailing zeros in $n!$ come only from factors of 10, i.e., pairs of 2 and 5 — and since factors of 2 are always more abundant than factors of 5 in a factorial, the zero count is simply the total power of 5 dividing $n!$, found by summing $\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$ until the term is 0.

- Trailing zeros in $100!$: $\text{floor}(100/5)=20$, $\text{floor}(100/25)=4$, $\text{floor}(100/125)=0$. Total = $20+4 = 24$ trailing zeros.

6. Base Conversion

Converting decimal to binary (or any base b) uses repeated division: divide by b , record the remainder, repeat on the quotient, and read the remainders bottom-to-top. Converting binary back to decimal is the reverse — multiply each bit by its positional power of 2 and sum.

- Convert 156 to binary: $156 \div 2 = 78$ r0, $78 \div 2 = 39$ r0, $39 \div 2 = 19$ r1, $19 \div 2 = 9$ r1, $9 \div 2 = 4$ r1, $4 \div 2 = 2$ r0, $2 \div 2 = 1$ r0, $1 \div 2 = 0$ r1. Reading remainders bottom-up: 10011100. Check: $128+16+8+4=156$.

7. Number of Factors and Sum of Factors

For $N = p^a \times q^b \times r^c$ (prime factorization), the total count of factors is $(a+1)(b+1)(c+1)$, and the sum of all factors is the product of the geometric series for each prime: $(1+p+p^2+\dots+p^a)(1+q+\dots+q^b)(1+r+\dots+r^c)$. This is a fast, purely formulaic technique that avoids listing factors by hand.

• $180 = 2^2 \times 3^2 \times 5^1$. Number of factors = $(2+1)(2+1)(1+1) = 3 \times 3 \times 2 = 18$. Sum of factors = $(1+2+4)(1+3+9)(1+5) = 7 \times 13 \times 6 = 546$.

Common Pitfalls

- Applying the divisibility-by-4 rule (last 2 digits) to check divisibility by 8, forgetting it needs the last 3 digits.
- Confusing HCF and LCM word problems — 'find when events coincide' always needs LCM, 'find the largest measure that divides evenly' always needs HCF.
- Forgetting the $\text{HCF} \times \text{LCM} = \text{product}$ identity, which is the fastest way to sanity-check an answer or solve for a missing number.
- Using the wrong cyclicity length — forgetting that when the exponent is exactly divisible by the cycle length, you use the LAST term of the cycle, not a 'remainder 0' term.
- Miscounting trailing zeros by including powers of 2 as well as 5 — only $\text{floor}(n/5) + \text{floor}(n/25) + \dots$ matters since 2s are never the bottleneck.
- In base conversion, reading the remainders top-to-bottom instead of bottom-to-top, which reverses the answer.

Key Takeaways

- Memorize divisibility rules for 2, 3, 4, 5, 8, 9, and 11 — they replace long division with a 5-second check.
- $\text{HCF} \times \text{LCM} = \text{product}$ of the two numbers; LCM solves 'coincide/repeat together' problems, HCF solves 'largest common measure' problems.
- Reduce huge exponents using cyclicity: find the repeat period of unit digits or remainders, then use exponent mod period.
- Trailing zeros in $n!$ = $\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$ (sum until the term becomes 0).
- Number of factors of $p^a \cdot q^b \cdot \dots$ is $(a+1)(b+1)\dots$; sum of factors is the product of each prime's geometric series.
- For base conversion, repeated division and reading remainders bottom-up is faster and less error-prone than trial and error.

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